COM 5506
Social Network Analysis for Communication

L3 Random Graphs

SEM B 2025/2026

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Review of L1 L2 key concepts

- Node, edges with features
- Degree (in/out), degree distribution
- Distance, average path length, network density
- Components, weakly/strongly connected
- Degree centrality, betweenness centrality, closeness centrality, eigenvalue centrality,

Today

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- Random graph model network
- Lab 2

We need to study random network models first

WHY?

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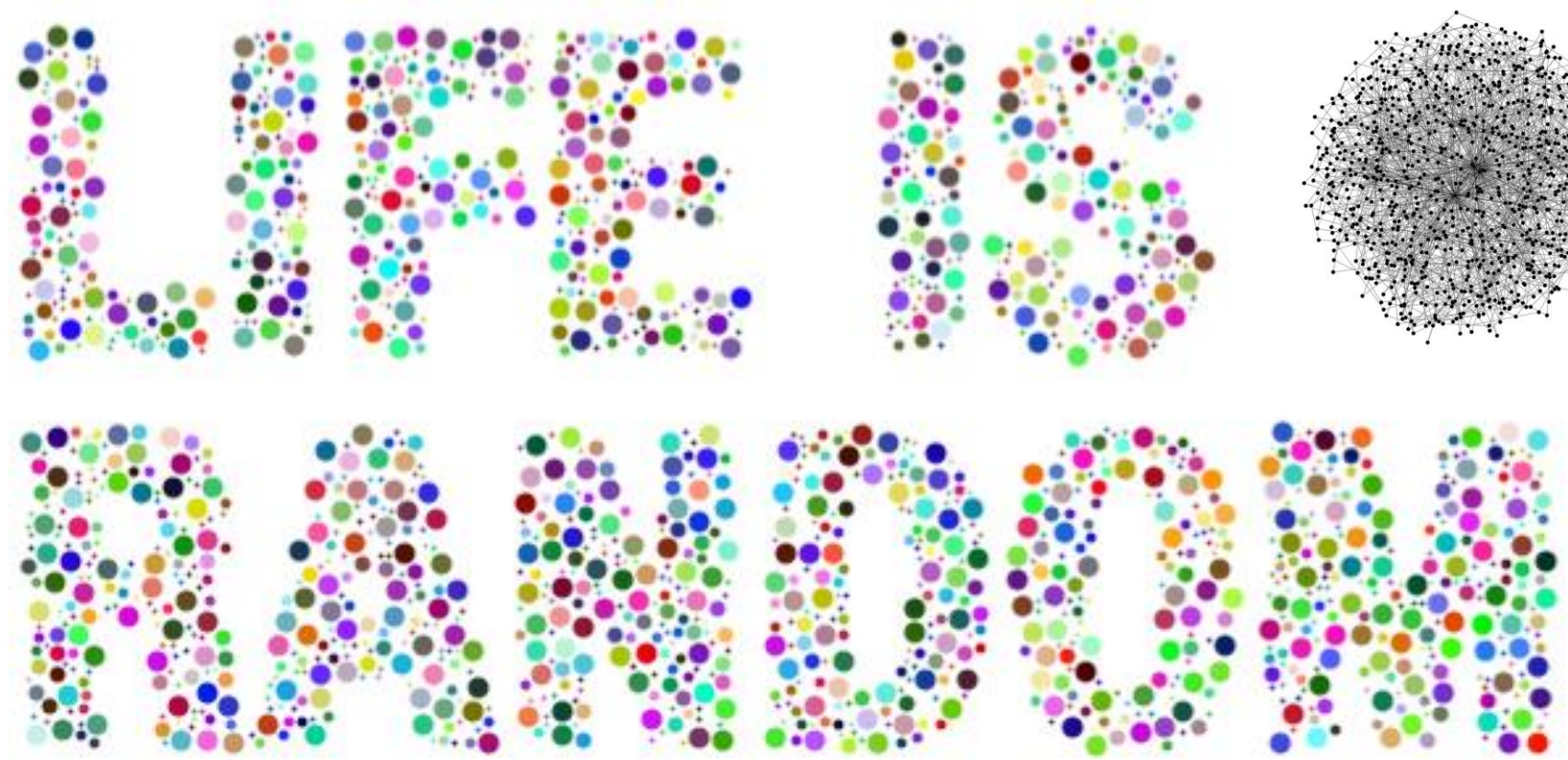

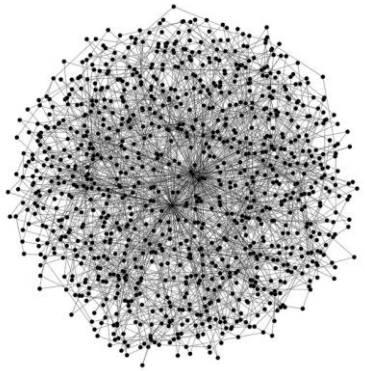

Why random?

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- Random means the simplest representation of complex reality (not necessarily true)
- Random means the purest status of an entity
- Randomness can help derive properties mathematically
- Randomness can help us predict outcomes

Random graph model

- Erdős–Rényi model: the simplest network model
  - Nodes connected at random
  - Network is not directed
- Key parameters (p or M, N)
  - p – probability that any two nodes have an edges
  - M – total number of edges in the graph
  - N – size of the network

Degree distribution

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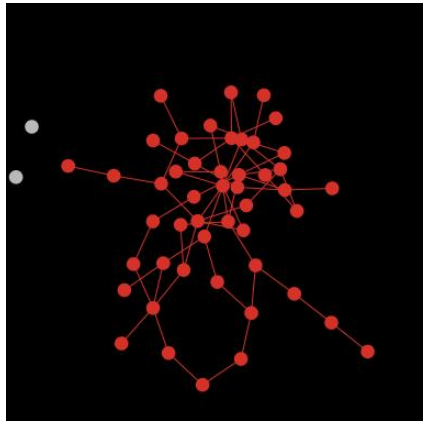
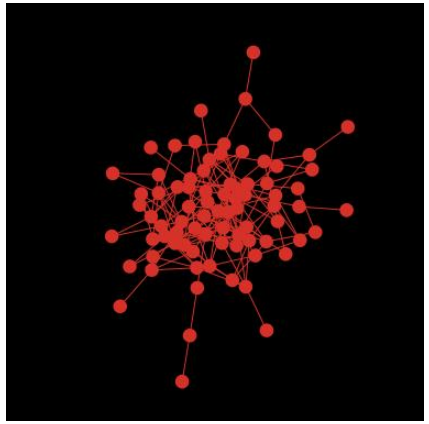
- (N, p) model: For each potential edge, flip a coin
  - the probability of adding the edge is p
  - The probability of having no edge is ____ ?
- As the size of the network increase, if you keep p, the probability of any two nodes being connected the same, what happens to the average degree (or mean degree)?

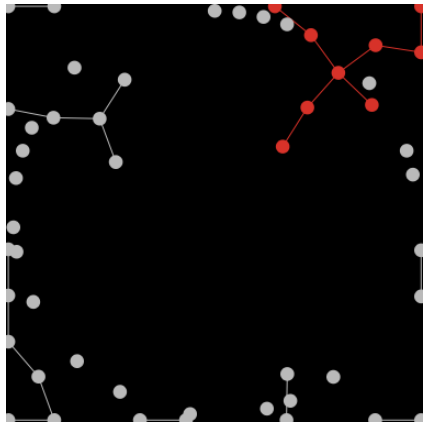
Examples: visual understanding



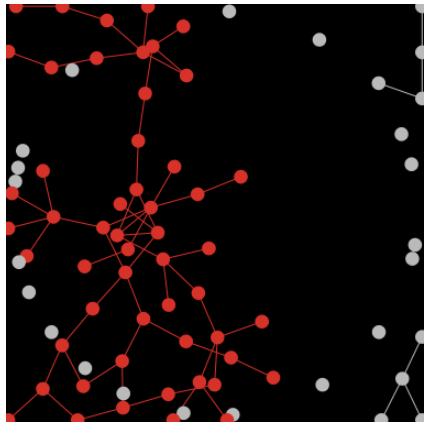
Avg degree: 2.4
N= 50
p= 0.05



Avg degree: 3.8
N= 80
p= 0.05



Avg degree: 0.88
N= 50
p= 0.02



Avg degree: 1.58
N= 80
p= 0.02

Mathematical insights

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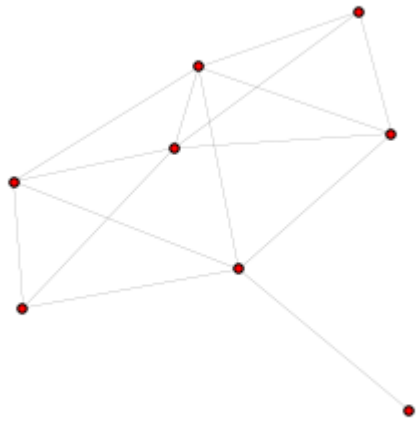
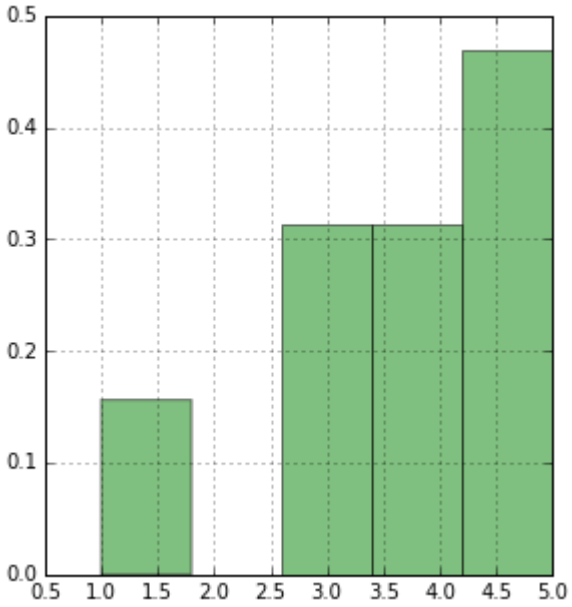
- Average degree: sum of degree divided by n
- Potential maximum degree proportional to n*(n-1) (divided by 2 if undirected)
- Potential degree in E-Y model: p*N*(n-1)
- Average degree (z) proportional to p*(n-1)
  - Holding p constant, AVD increases as N increase
  - Holding N constant, AVD increases as p increase



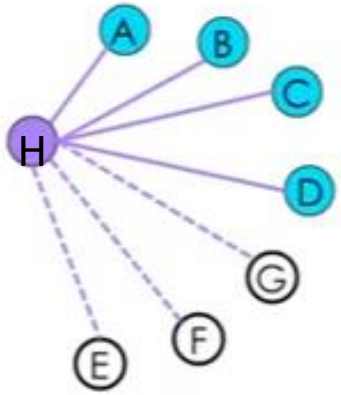


We have an 8 node graph, the probability p of any two nodes being connected is 0.5.

What is the probability that a given node has 4 edges?

basic stats review

- We have an 8 node graph, the probability p of any two nodes being connected is 0.5.
- What is the probability that a given node has 4 edges?



number of ways of choosing k items out of (n-1)

$$= \frac{\text{number of ways of arranging } \boldsymbol{n\text{-}1} \text{ items}}{(\text{\# of ways to arrange } \boldsymbol{k} \text{ things})*(\text{\# ways to arrange } \boldsymbol{n}\text{-1-}\boldsymbol{k} \text{ things})}$$

$$= \frac{\boldsymbol{n\text{-}1}!}{\boldsymbol{k}!\,(\boldsymbol{n}\text{-1-}\boldsymbol{k})!}$$

$$B(7;4;p) = \binom{7}{4} p^4 (1-p)^3$$

How many edges per node?

- The binominal distribution gives us the probability that a node has degree k:

k times of succeeded link

$$B(N-1;k;p) = \binom{N-1}{k} p^k (1-p)^{N-1-k}$$

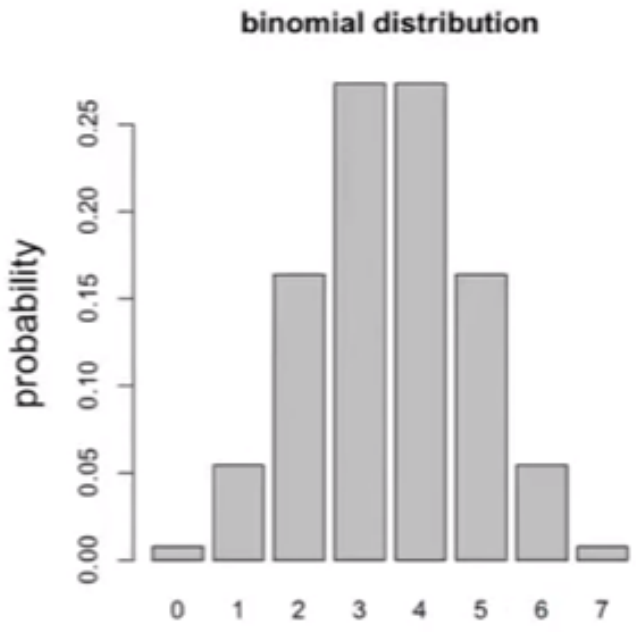
N-1-k times of no link

binominal coefficient

p and shape

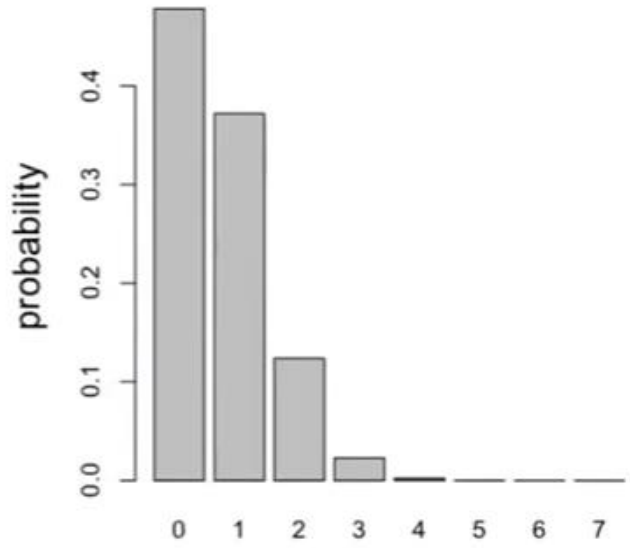
- No hubs in random graph model
- Probabilities sum to 1

p=0.5



mean degree = (n-1)*p = 3.5

p=0.1



mean degree = (n-1)*p = 0.7

Exercise:

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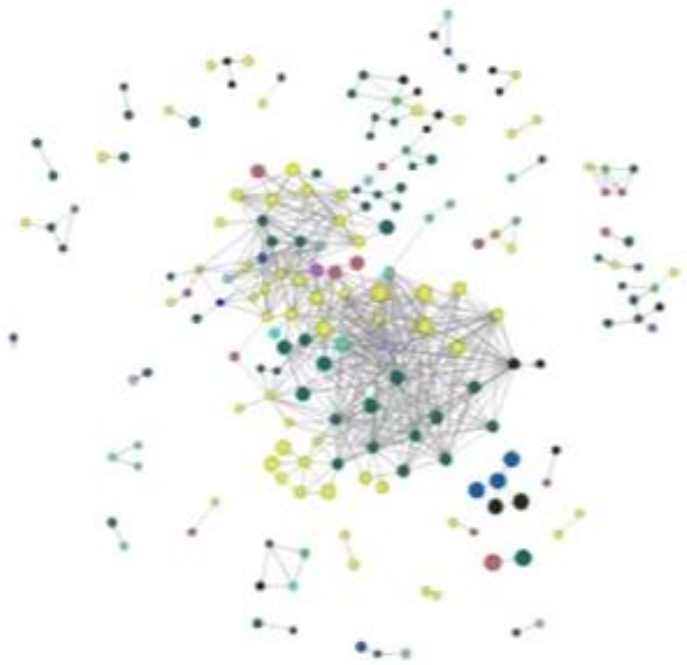
- What is the average degree of a graph with 10 nodes and probability p=1/3 of an edge existing between any two nodes?
  - 1
  - 2
  - 3
  - 4

Question:

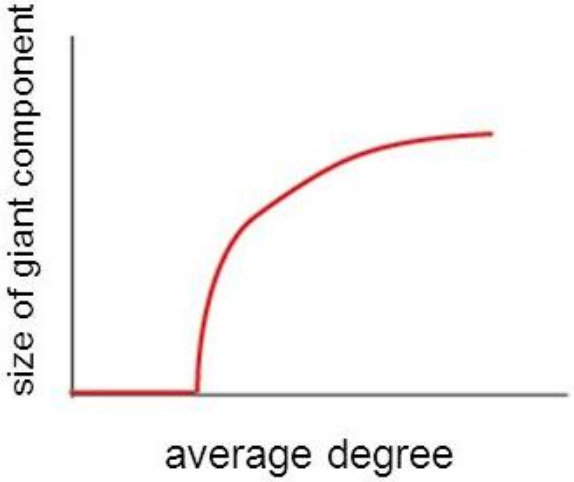
- What is the average degree z at which a giant component starts to emerge? 0, 1, 3/2, 3 ?

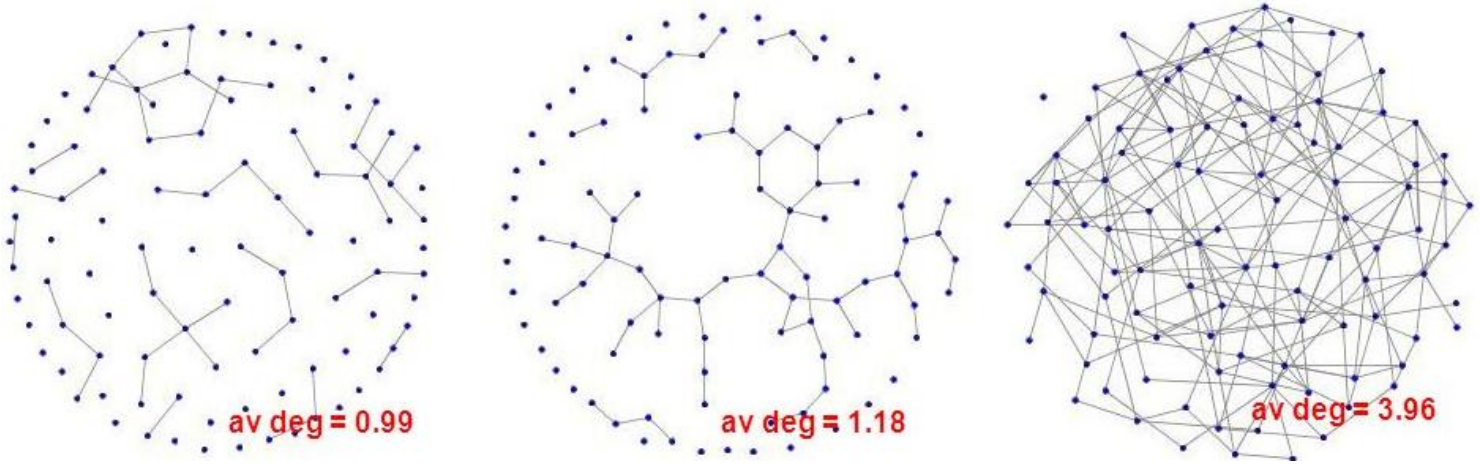
- If the largest component encompasses a significant fraction of the graph, it is called the giant component.



When will giant component emerge?

- As the average degree increases to z=1, a giant component suddenly appears.
- By property of degree distribution, at z=1, each of your friend is expected to have another friend, who in turn have another friend…





When will giant component emerge?

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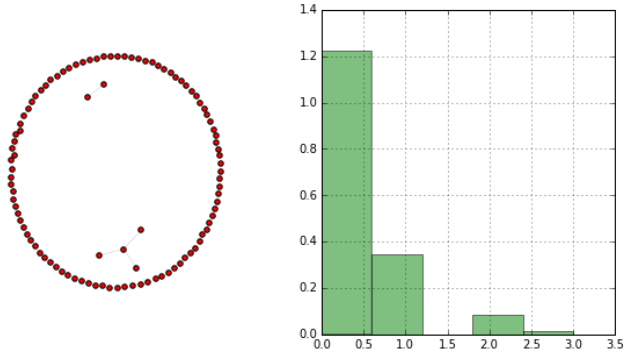
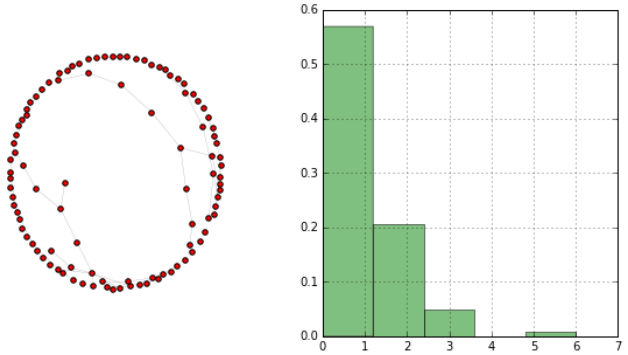
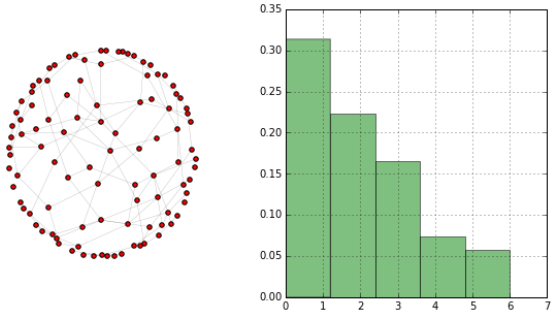
- Remember the expected degree, or the mean degree of a random graph is E[X] = z = (n-1)p
- If z=1, then p = 1/(n-1)
- In other words, when the probability of the two nodes having an edge reaches the reverse of the size of the network minus one, then a giant component will appear.



Z = 0.5



Z = 1



Z = 2

When will isolated nodes disappear?

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- p(network has degree 0) = $(1-p)^{n-1}$, so when $(1-p)^{n-1} = 0$, there is no isolated nodes.
- This answer is not very informative, because two parameters p and n determines the probability, and we don't know either one; and p and n are independent.
- This formula does not reveal the inner relationship between the two variables when they satisfy the condition that network has no isolated nodes.
- So we wish to control for one parameter. How can we construct the network so that the ***mean degree will not vary with (or independent of) the size of the network***?
- Let p = c/(n-1), c being a constant

When will isolated nodes disappear?

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- Derive the probability: a single node is isolated

$$P[v \text{ has degree } 0] = (1-p)^{n-1} = \left(1 - \frac{c}{n-1}\right)^{n-1} \xrightarrow[n\to\infty]{} e^{-c}$$

$$\lim_{n\to\infty}\left(1-\frac{c}{n-1}\right)^{n-1} = \left(1-\frac{1}{x}\right)^{-x\cdot c} = \left[\underbrace{\lim_{x\to\infty}\left(1-\frac{1}{x}\right)^{-x}}_{e \text{ (by definition)}}\right]^{-c} = e^{-c}$$

Use substitution $\frac{1}{x} = \frac{c}{n-1}$

When will isolated nodes disappear?

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- At least one isolated node p'(isolated) = $n*e^{-c}$
- We want this p' = 1, so how?
- Let us try:
  - If c = ln(n), then p'(isolated) = $n*e^{-\ln(n)} = n*(1/n)=1$
  - If c= 2ln(n), then p'(isolated) = $n*e^{-2\ln(n)} = n*(1/n^2) = 1/n \rightarrow 0$, as $n \rightarrow \infty$
- Since we make p=c/(n-1)
- So if:
  - p= ln(n)/(n-1) then p'(isolated) =1
  - p= 2ln(n)/(n-1) then p'(isolated)=0

Pulling everything together

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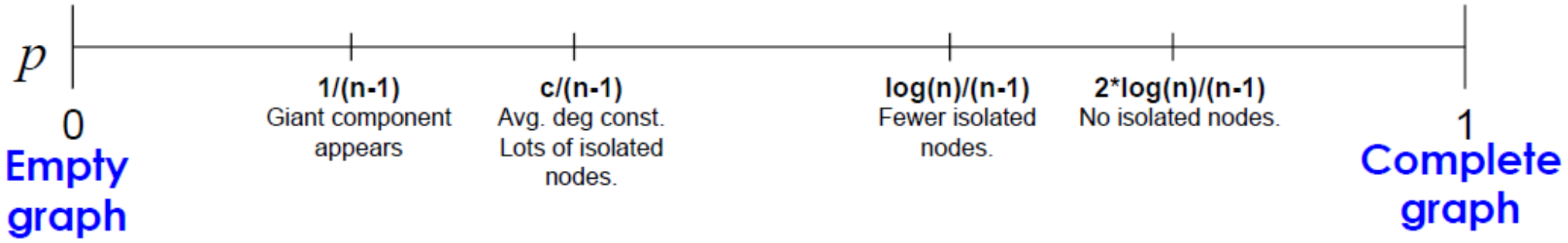
- The graph structure of G_{np} changes along with the size of *p*



Lab 2

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- Network graph further
- Exercise